



COM3021/COMM115

Data Science at Scale

Introduction

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Agenda

- Introduction

The emergence of the big data era

Applications

- Challenges in large-scale data analytics
-

Introduction

- Over the last decade we have witnessed a dramatic change in how companies do business becoming increasingly:

Social

Customer-oriented

Dynamic

- Not only tech companies
-

How?

- Incorporated into their workflows tasks such as

Monitoring

Collecting data

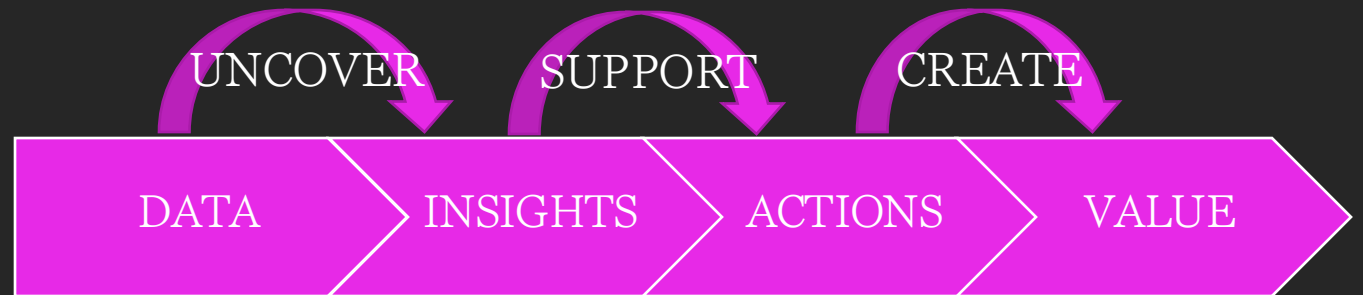
Understanding

Learning from the data

Responding

Improving

Adapting



A DAY IN DATA

The exponential growth of data is undisputed, but the numbers behind this explosion – fuelled by internet of things and the use of connected devices – are hard to comprehend, particularly when looked at in the context of one day

500m

tweets are sent every day
Twitter

294bn

billion emails are sent
Radicati Group

320bn

emails to be sent each day by 2021

306bn

emails to be sent each day by 2020

3.9bn

people use emails

4PB

of data created by Facebook, including

350m photos

100m hours of video watch time
Facebook Research

4TB

of data produced by a connected car
Intel

ACCUMULATED DIGITAL UNIVERSE OF DATA

4.4ZB

2019

44ZB

2020
PwC

DEMYSTIFYING DATA UNITS

From the more familiar 'bit' or 'megabyte', larger units of measurement are more frequently being used to explain the masses of data

Unit	Value	Size
b	bit	0 or 1
B	byte	8 bits
KB	kilobyte	1,000 bytes
MB	megabyte	1,000 ³ bytes
GB	gigabyte	1,000 ³ bytes
TB	terabyte	1,000 ³ bytes
PB	petabyte	1,000 ³ bytes
EB	exabyte	1,000 ³ bytes
ZB	zettabyte	1,000 ³ bytes
YB	yottabyte	1,000 ³ bytes

*Lowercase "b" is used as an abbreviation for bit, while an uppercase "B" represents bytes.

65bn

messages sent over WhatsApp and two billion minutes of voice and video calls made
Facebook

Searches made a day **5bn**

Searches made a day from Google **3.5bn**

463EB

of data will be created every day by 2025
IDC

95m

photos and videos are shared on Instagram
Instagram Business

28PB

to be generated from wearable devices by 2020
Statista

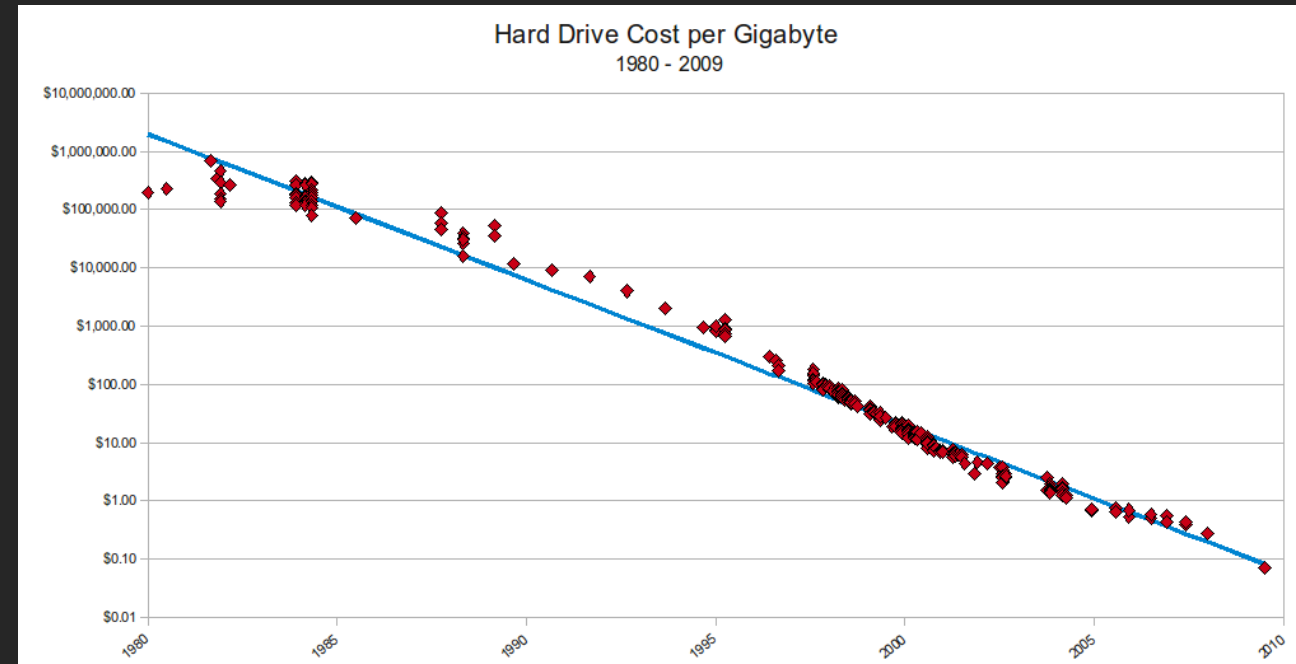
RACONTEUR

463 EB in 2021 Estimate*

495.9 EB (Millions TB) in 2025 Estimates**

How?

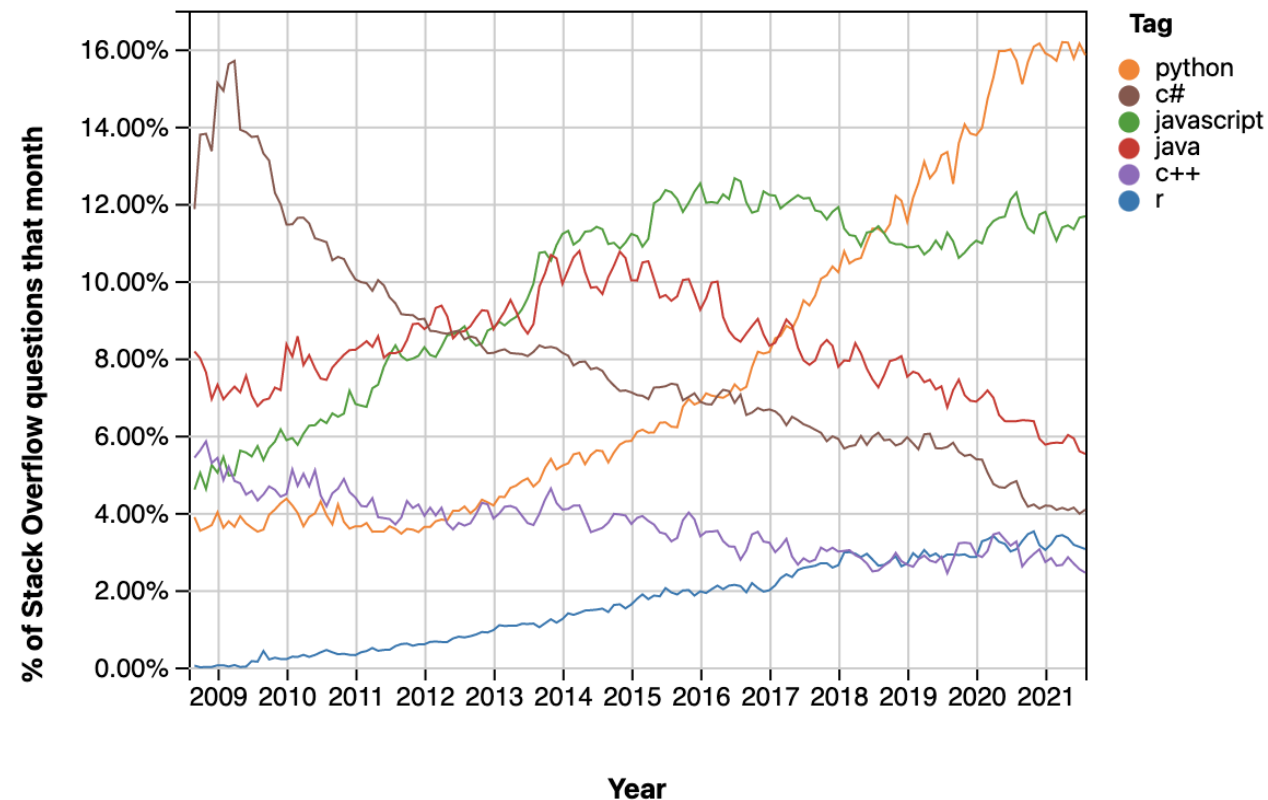
- Cheap storage and processing
- Free, open-source tools
- Faster networks



Source: Steyn, Jacques & Kirlidog, Melih. (2013). A BRIEF CRITIQUE OF THE DOMINANT APPROACHES TO ICT4D. A CONTEXTUALIST APPROACH. A BRIEF CRITIQUE OF THE DOMINANT APPROACHES TO ICT4D. A CONTEXTUALIST APPROACH.

How?

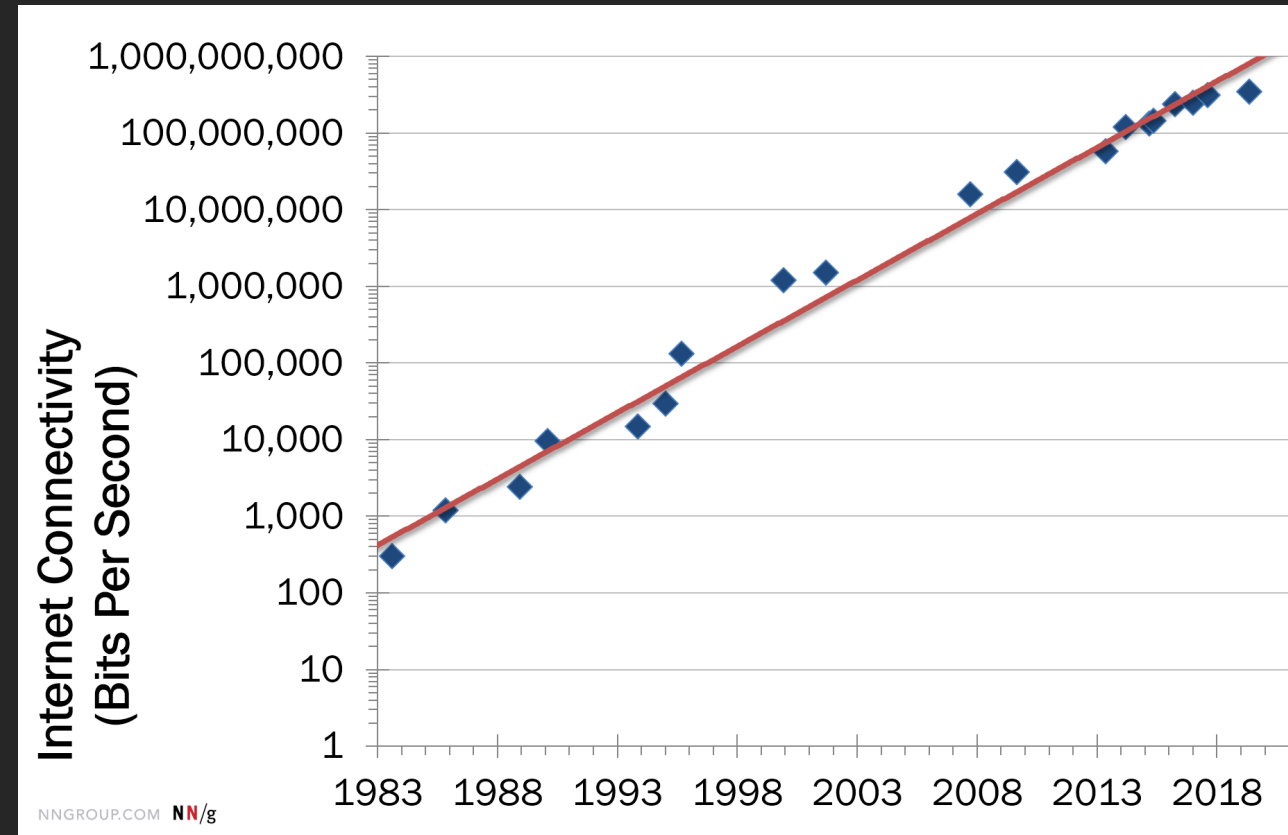
- Cheap storage and processing
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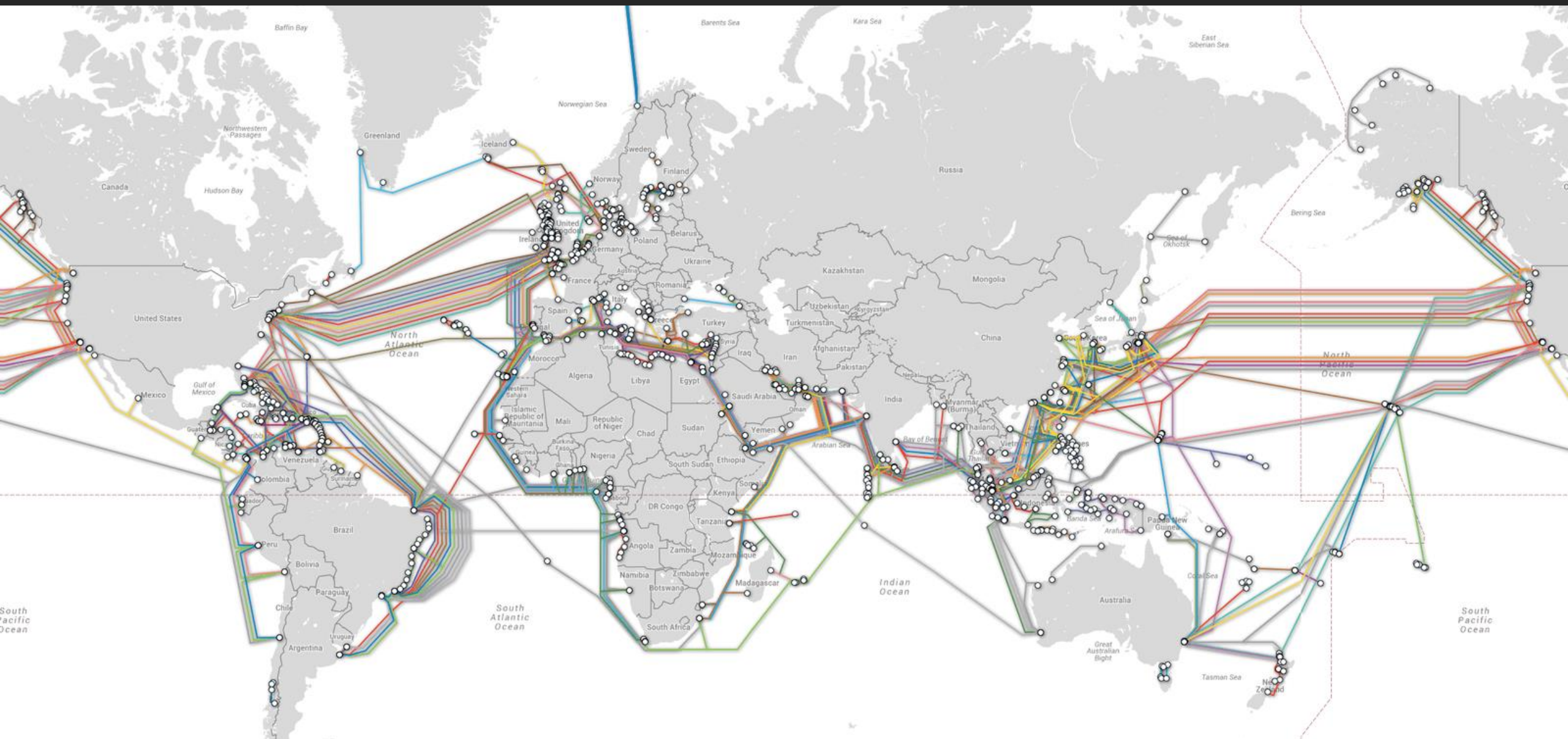
Source: StackOverflow Trends

How?

- Cheap storage and processing
- Free, open-source tools
- Faster networks



Source: <https://www.nngroup.com/articles/law-of-bandwidth/>





Costs

- Internet companies (Google, Amazon, Microsoft) are pushing the boundaries
Developing new solutions to handle massive amounts of data more efficiently
Cloud-based, infrastructure as a Service (IaaS) such as GCP and AWS
-

Big Data era

- Massive amounts of data
 - +
 - Cheaper and more powerful data processing tools
 - =
 - Better models
 - Generalisation, prediction, inference*
 - Beyond better solutions to existing problems
-

The data sources are changing

- Before 1980
Data was generated by machines
 - From 1980 to 2000
Data was being generated by employees resulting from processes within organisations
Payroll, bank transactions, sales
 - 2000~2005
Data was generated by the end users
Videos, blog posts, photos
 - From ~2005
Data also started to be generated by devices, hardware and applications
Mobile phones, remote sensors, logs
-

Examples

- IoT includes mobile devices, microphones, readers/sensors/scanners that generate data.
- CCTV for security and military surveillance, animal tracker, etc.
- Call logs analytics in the customer service centre.
- An aircraft can generate several terabytes of data per flight.
- Smart meters in power grids read the usage every 15 minutes and records 350 billion transactions in a year.

Areas transformed by the
data revolution

Artificial Intelligence



Marketing

- Companies like Walmart, Netflix and Amazon
Customer Profiling
 - Targeted advertising
 - Personalised communication
 - Better recommendations
-

HEALTHCARE

Personalised treatments

Drug discovery

Find cause of diseases

Cancer treatment algorithm developed by UH Cancer Center

[News](#) » [Research](#) » Cancer treatment algorithm developed...

August 9, 2016 [UH News](#)



Gordon Okimoto

[University of Hawai'i Cancer Center](#) researchers developed a computational algorithm to analyze "big data" obtained from tumor samples to better understand and treat cancer.



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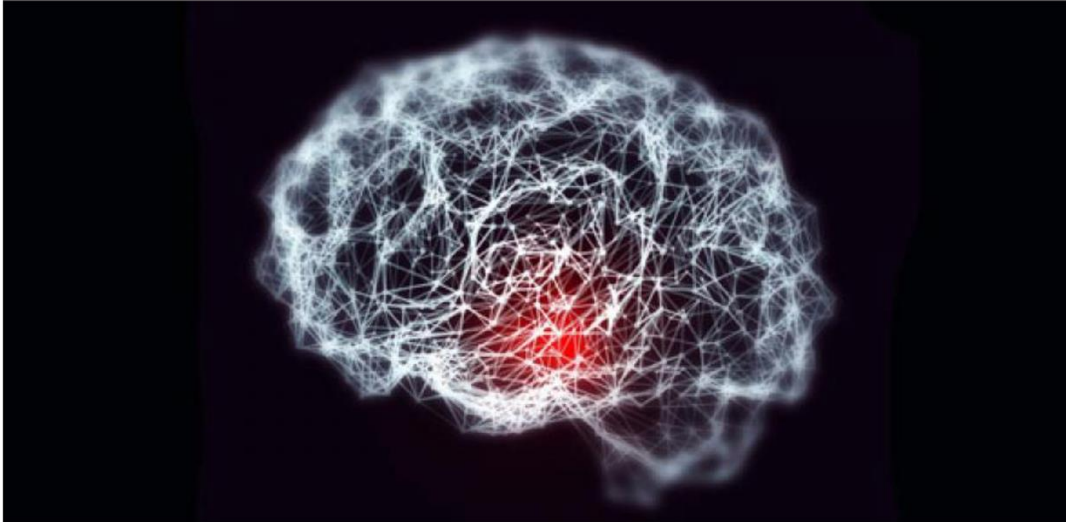
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Scientists reveal plan to target the cause of Alzheimer's disease



Researchers have developed a new way to target the toxic particles that destroy healthy brain cells in Alzheimer's disease.

SMART CITIES

Optimising public transport
schedules

Traffic modelling and forecasting

Intelligent traffic systems

Self-driving vehicles



DISASTER RESPONSE

Optimising evacuation routes

Monitoring human rights
violation

Optimising resource allocation



Analysing large scale data

Challenges

The 3Vs of big data

- Volume

We are generating data at an unprecedented rate

163 zettabytes of data created up to 2020

3000% increase in 16 years

Datasets are so large they cannot be analysed on single machines

- Velocity

The rate they arrive imposes will require specialised tools

Storage

Models

- Variety

Unstructured, multi-modal

Before we had tabular data in spreadsheets

Now we have text, audio, video

There are more Vs

- Veracity
- Value

